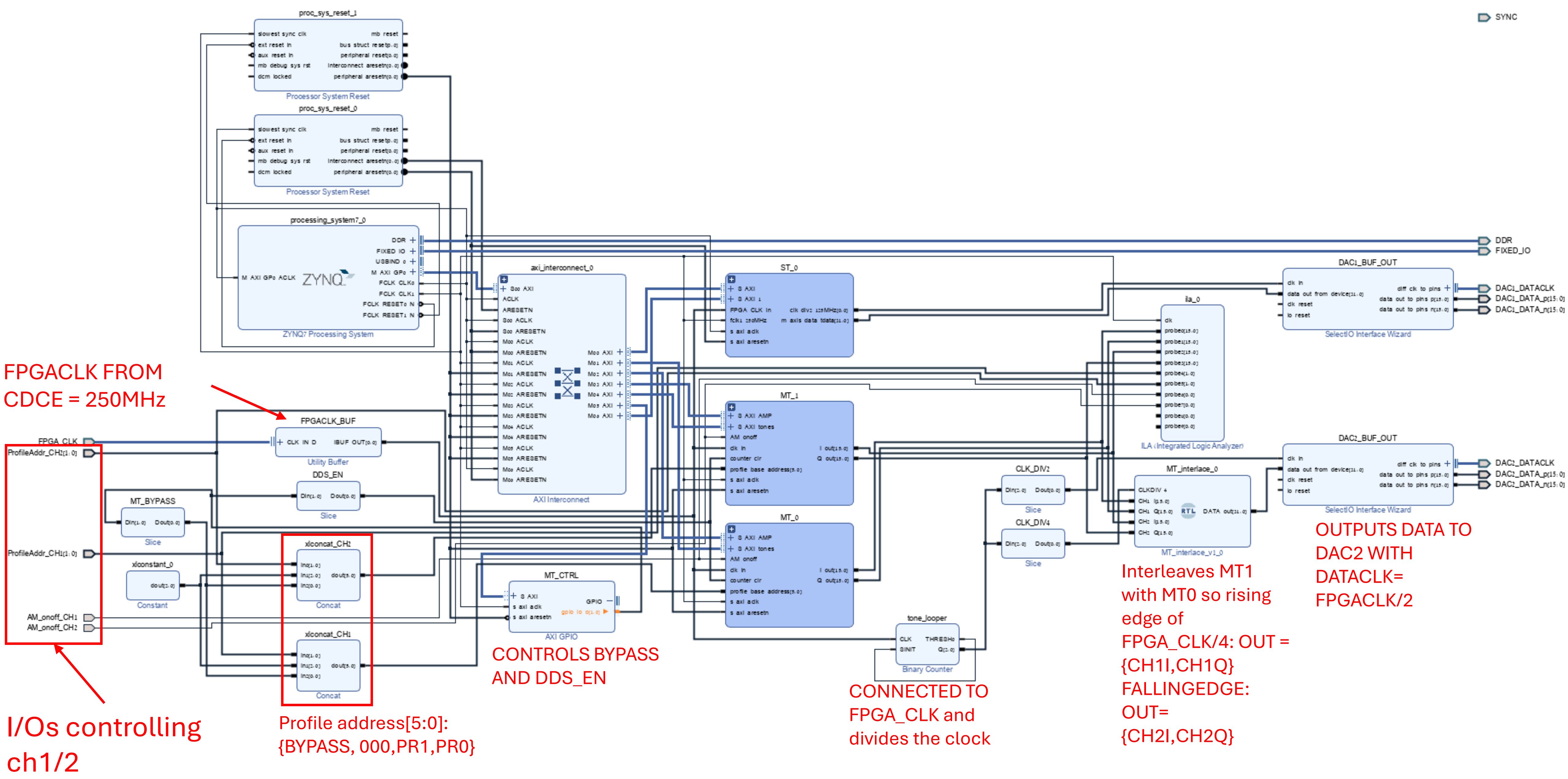
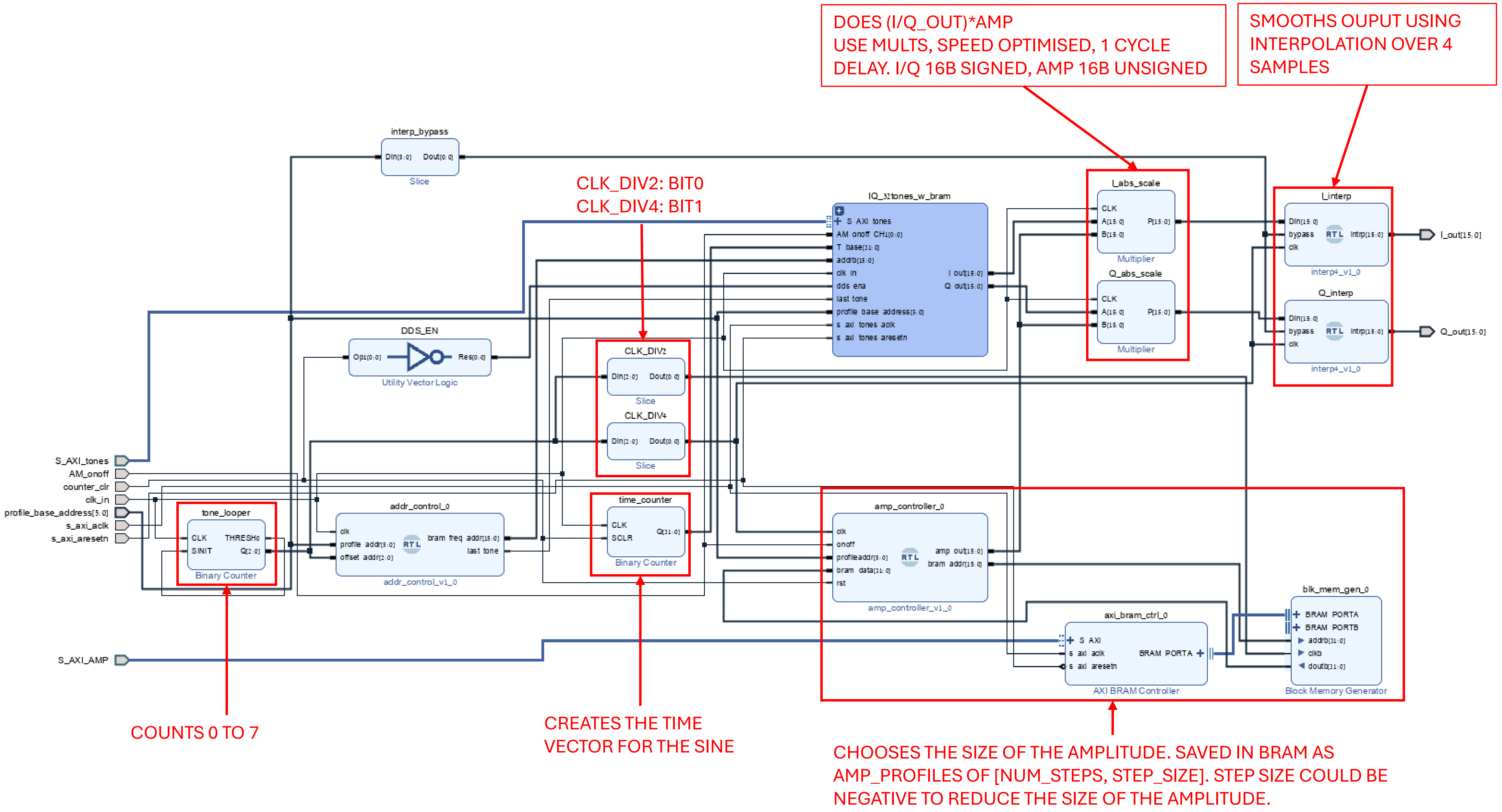
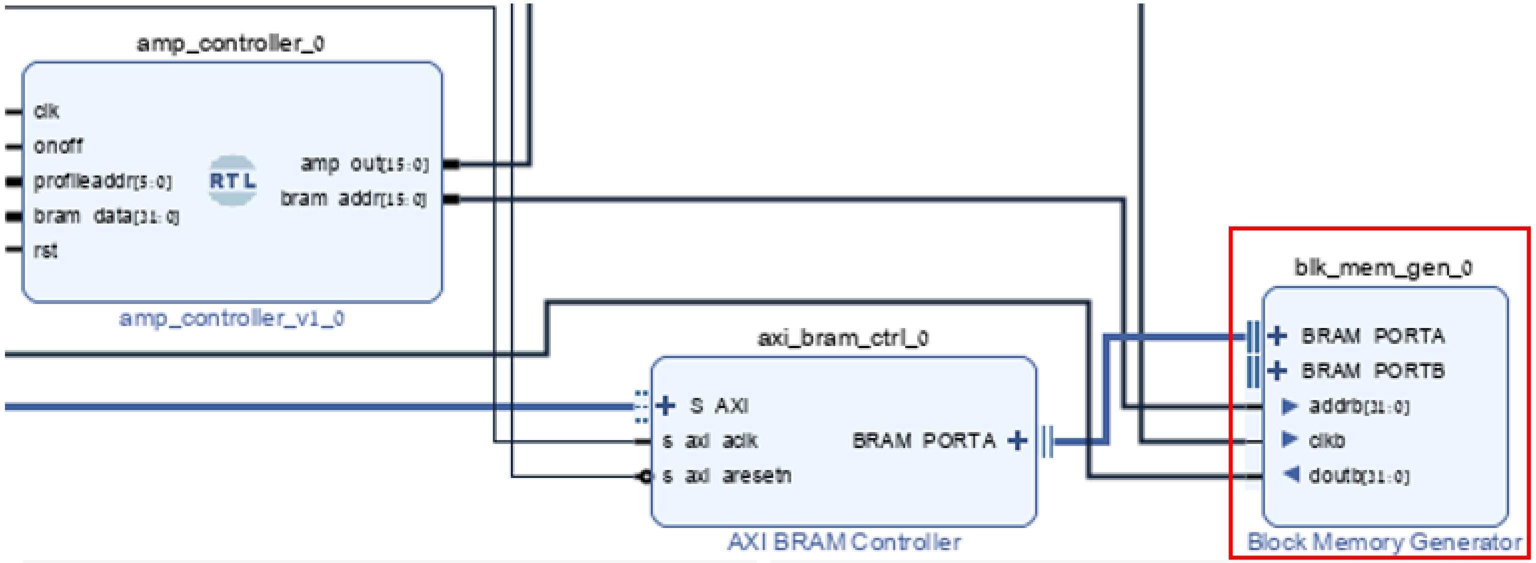
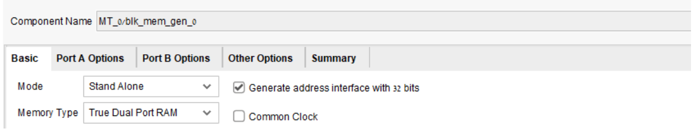
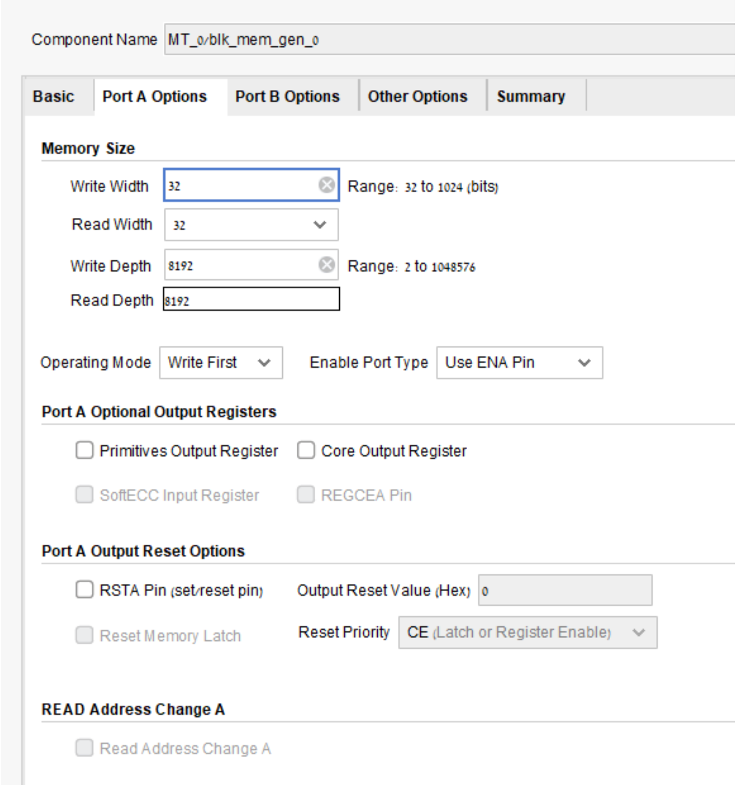
Multi Tone Vivado BD Breakdown

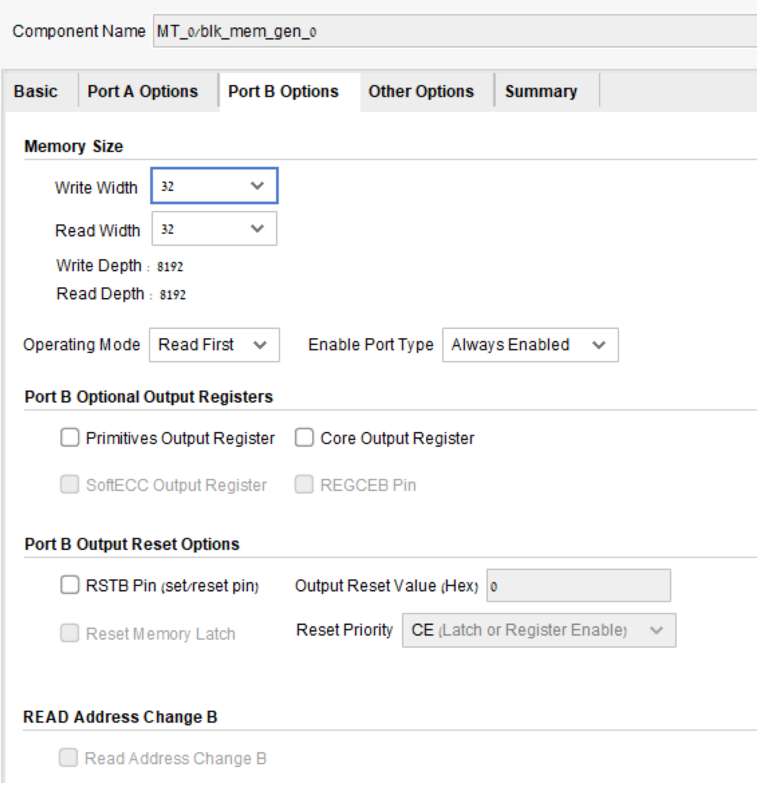

proc_sys_reset_1
Processor System Reset
proc_sys_reset_0
Processor System Reset
processing_system7_0
ZYNQ
ZYNQ7 Processing System
axi_interconnect_0
AXI Interconnect
ST_0
MT_1
MT_0
ila_0
ILA (Integrated Logic Analyzer)
DAC1_BUF_OUT
SelectIO Interface Wizard
DAC2_BUF_OUT
SelectIO Interface Wizard
MT_interlace_0
MT_interlace_v1_0
CLK_DIV2
Slice
CLK_DIV4
Slice
tone_looper
Binary Counter
MT_CTRL
AXI GPIO
FPGACLK_BUF
Utility Buffer
DDS_EN
Slice
MT_BYPASS
Slice
xlconstant_0
Constant
xlconcat_CH2
Concat
xlconcat_CH1
Concat
FPGA_CLK
ProfileAddr_CH2[1:0]
ProfileAddr_CH1[1:0]
AM_onoff_CH1
AM_onoff_CH2
SYNC
DDR
FIXED_IO
DAC1_DATACLK
DAC1_DATA_p[15:0]
DAC1_DATA_n[15:0]
DAC2_DATACLK
DAC2_DATA_p[15:0]
DAC2_DATA_n[15:0]
FPGACLK FROM CDCE = 250MHz
I/Os controlling ch1/2
Profile address[5:0]: {BYPASS, 000,PR1,PR0}
CONTROLS BYPASS AND DDS_EN
CONNECTED TO FPGA_CLK and divides the clock
Interleaves MT1 with MT0 so rising edge of FPGA_CLK/4: OUT = {CH1I,CH1Q} FALLINGEDGE: OUT= {CH2I,CH2Q}
OUTPUTS DATA TO DAC2 WITH DATACLK= FPGACLK/2

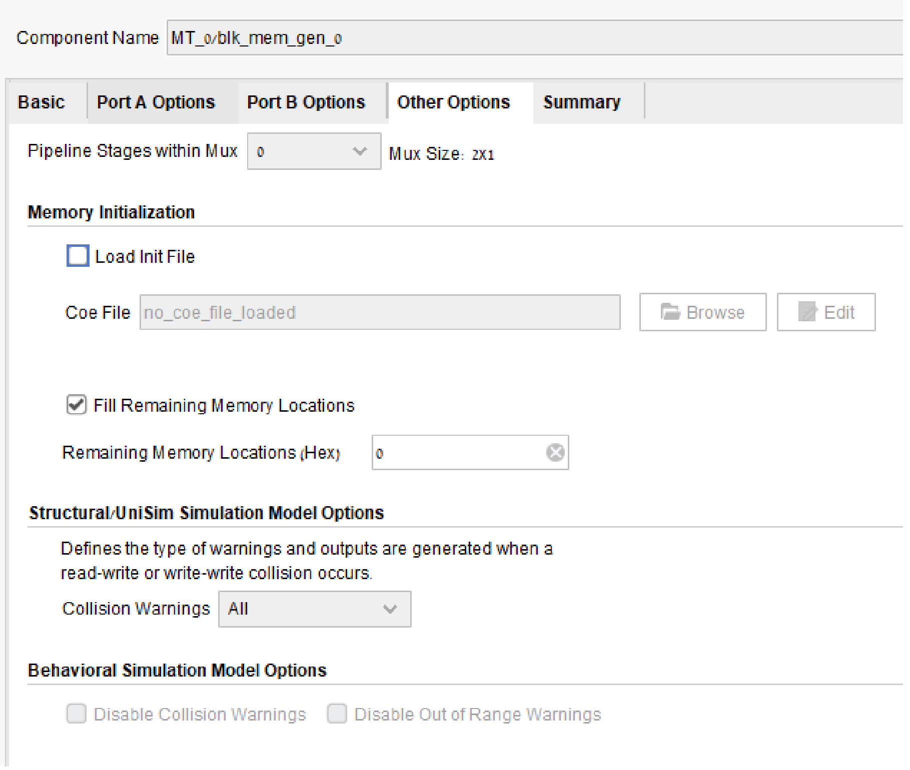

DOES (I/Q_OUT)*AMP
USE MULTS, SPEED OPTIMISED, 1 CYCLE DELAY. I/Q 16B SIGNED, AMP 16B UNSIGNED
SMOOTHS OUPUT USING INTERPOLATION OVER 4 SAMPLES
CLK_DIV2: BIT0
CLK_DIV4: BIT1
interp_bypass
Slice
IQ_32tones_w_bram
I_abs_scale
Multiplier
Q_abs_scale
Multiplier
I_interp
interp4_v1_0
Q_interp
interp4_v1_0
I_out[15:0]
Q_out[15:0]
DDS_EN
Utility Vector Logic
CLK_DIV2
Slice
CLK_DIV4
Slice
S_AXI_tones
AM_onoff
counter_clr
clk_in
profile_base_address[5:0]
s_axi_aclk
s_axi_aresetn
tone_looper
Binary Counter
addr_control_0
addr_control_v1_0
time_counter
Binary Counter
amp_controller_0
amp_controller_v1_0
axi_bram_ctrl_0
AXI BRAM Controller
blk_mem_gen_0
Block Memory Generator
S_AXI_AMP
COUNTS 0 TO 7
CREATES THE TIME VECTOR FOR THE SINE
CHOOSES THE SIZE OF THE AMPLITUDE. SAVED IN BRAM AS AMP_PROFILES OF [NUM_STEPS, STEP_SIZE]. STEP SIZE COULD BE NEGATIVE TO REDUCE THE SIZE OF THE AMPLITUDE.

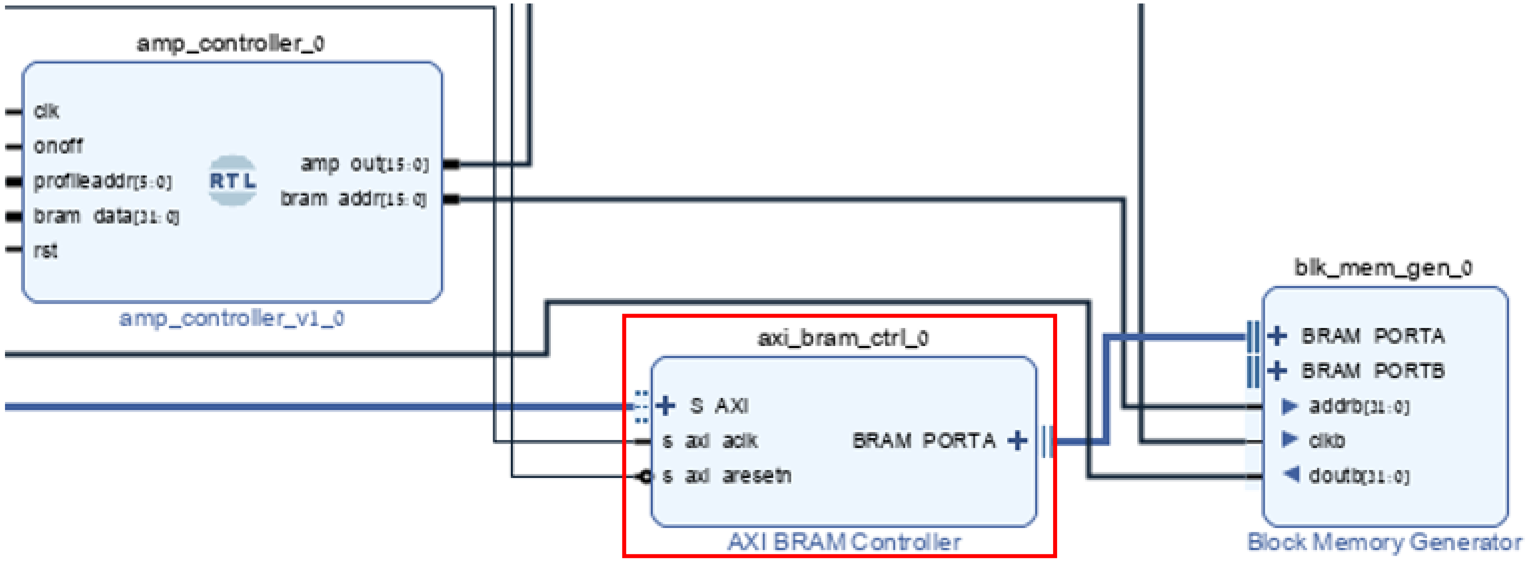

amp_controller_0
clk
onoff
profileaddr[5:0]
bram_data[31:0]
rst
RTL
amp_out[15:0]
bram_addr[15:0]
amp_controller_v1_0
axi_bram_ctrl_0
S_AXI
s_axi_aclk
s_axi_aresetn
BRAM_PORTA
AXI BRAM Controller
blk_mem_gen_0
BRAM_PORTA
BRAM_PORTB
addrb[31:0]
clkb
doutb[31:0]
Block Memory Generator

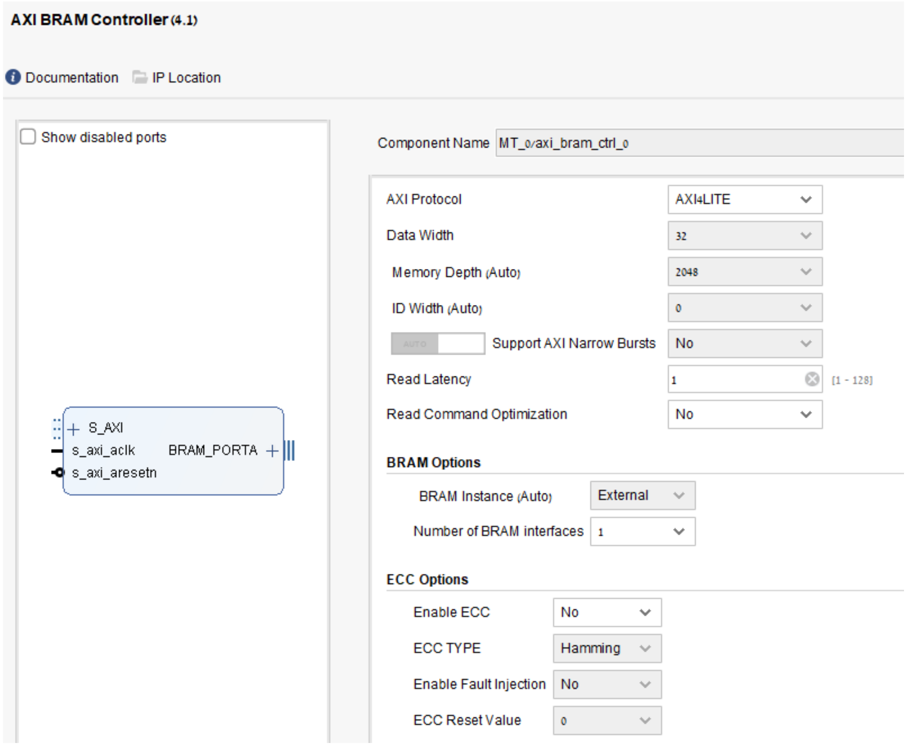

Component Name MT_0/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Mode Stand Alone
Generate address interface with 32 bits
Memory Type True Dual Port RAM
Common Clock

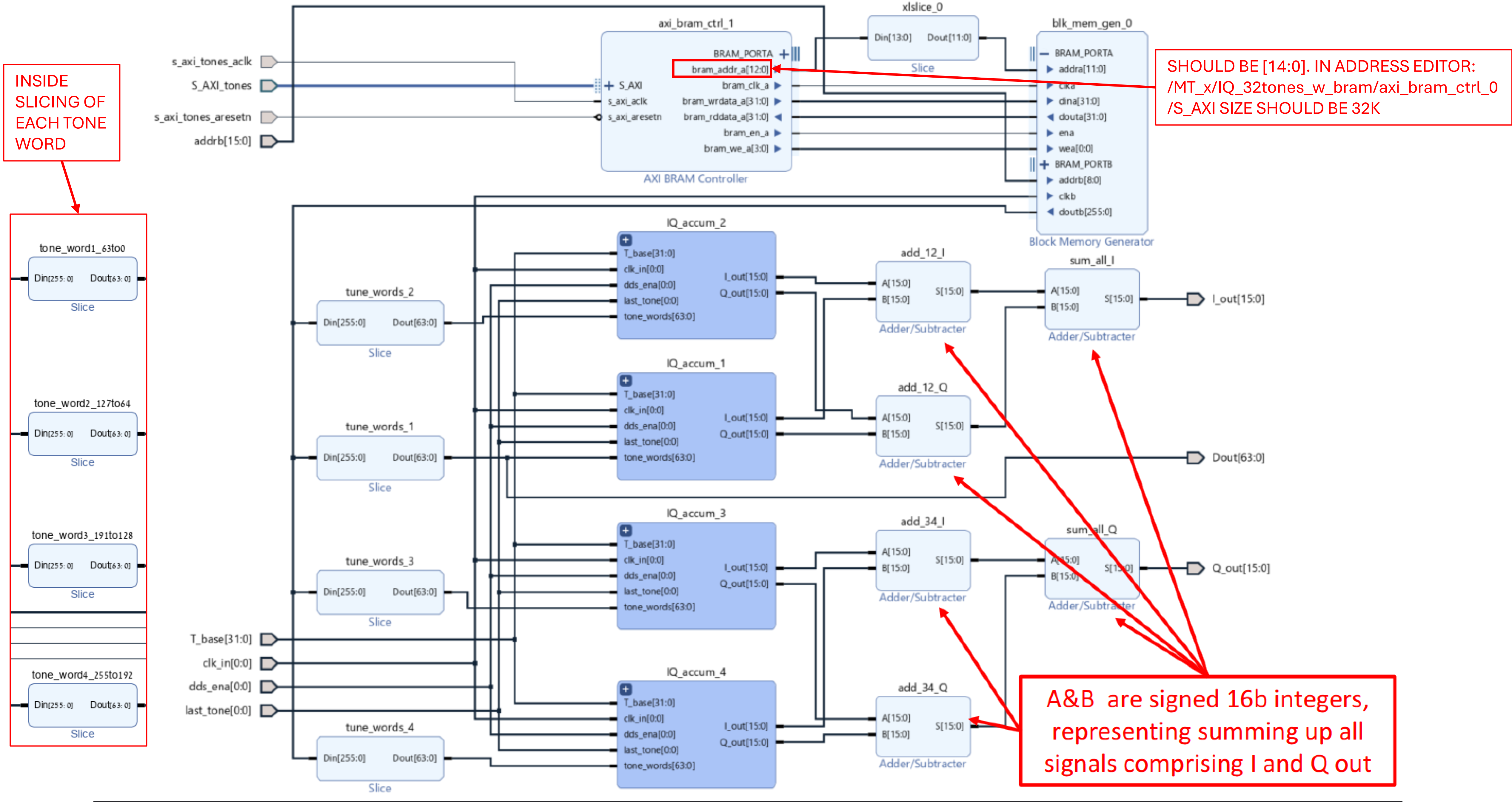

Component Name MT_0/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Memory Size
Write Width 32
Range: 32 to 1024 (bits)
Read Width 32
Write Depth 8192
Range: 2 to 1048576
Read Depth 8192
Operating Mode Write First
Enable Port Type Use ENA Pin
Port A Optional Output Registers
Primitives Output Register
Core Output Register
SoftECC Input Register
REGCEA Pin
Port A Output Reset Options
RSTA Pin (set/reset pin)
Output Reset Value (Hex) 0
Reset Memory Latch
Reset Priority CE (Latch or Register Enable)
READ Address Change A
Read Address Change A


Component Name MT_0/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Memory Size
Write Width 32
Read Width 32
Write Depth : 8192
Read Depth : 8192
Operating Mode Read First
Enable Port Type Always Enabled
Port B Optional Output Registers
Primitives Output Register
Core Output Register
SoftECC Output Register
REGCEB Pin
Port B Output Reset Options
RSTB Pin (set/reset pin)
Output Reset Value (Hex) 0
Reset Memory Latch
Reset Priority CE (Latch or Register Enable)
READ Address Change B
Read Address Change B

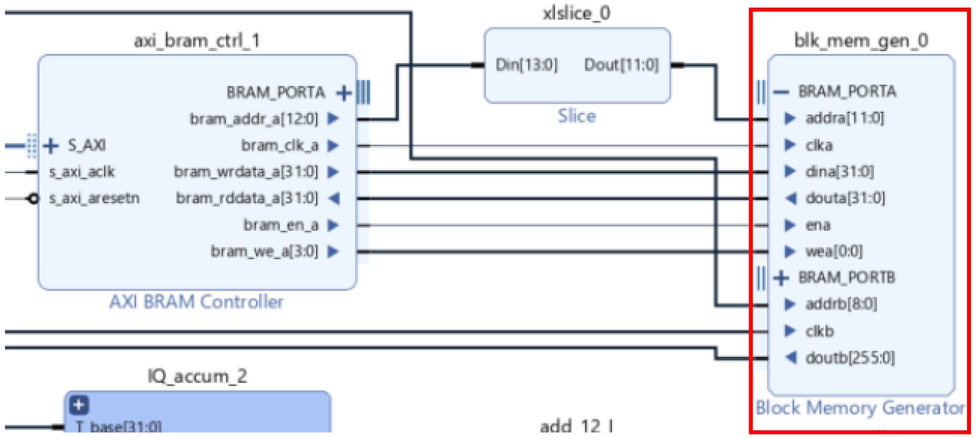

Component Name MT_0/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Pipeline Stages within Mux 0
Mux Size: 2x1
Memory Initialization
Load Init File
Coe File no_coe_file_loaded
Browse
Edit
Fill Remaining Memory Locations
Remaining Memory Locations (Hex) 0
Structural/UniSim Simulation Model Options
Defines the type of warnings and outputs are generated when a
read-write or write-write collision occurs.
Collision Warnings All
Behavioral Simulation Model Options
Disable Collision Warnings
Disable Out of Range Warnings

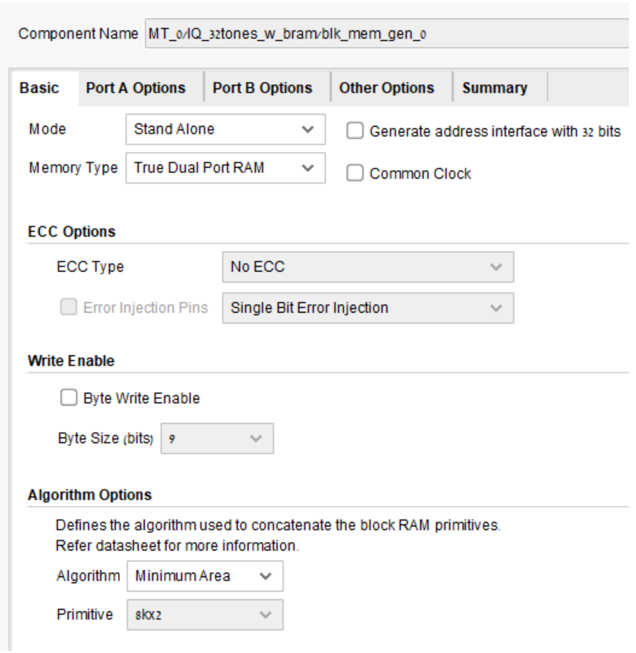

amp_controller_0
clk
onoff
profileaddr[5:0]
bram data[31:0]
rst
RTL
amp out[15:0]
bram addr[15:0]
amp_controller_v1_0
axi_bram_ctrl_0
S AXI
s axi aclk
s axi aresetn
BRAM PORTA
AXI BRAM Controller
blk_mem_gen_0
BRAM PORTA
BRAM PORTB
addrb[31:0]
clkb
doutb[31:0]
Block Memory Generator

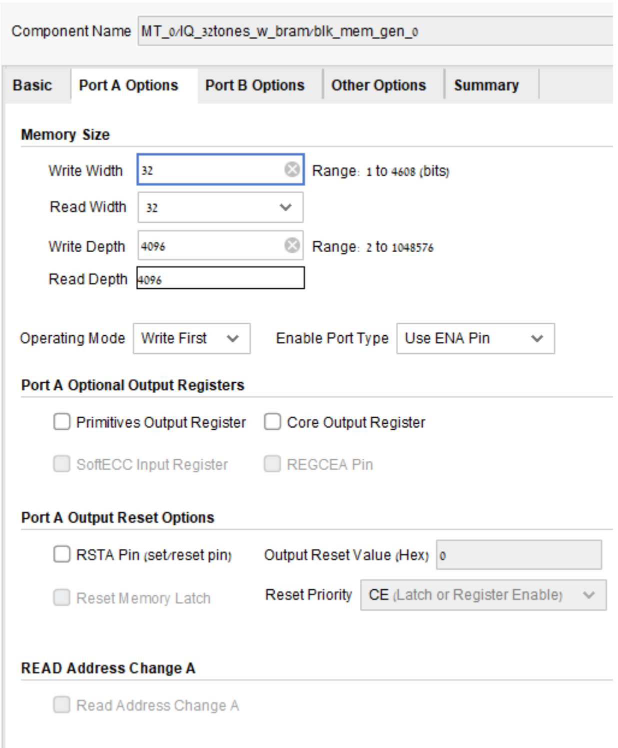

AXI BRAM Controller (4.1)
Documentation
IP Location
Show disabled ports
S_AXI
s_axi_aclk
BRAM_PORTA
s_axi_aresetn
Component Name
MT_0/axi_bram_ctrl_0
AXI Protocol
AXI4LITE
Data Width
32
Memory Depth (Auto)
2048
ID Width (Auto)
0
Support AXI Narrow Bursts
No
Read Latency
1
[1 - 128]
Read Command Optimization
No
BRAM Options
BRAM Instance (Auto)
External
Number of BRAM interfaces
1
ECC Options
Enable ECC
No
ECC TYPE
Hamming
Enable Fault Injection
No
ECC Reset Value
0

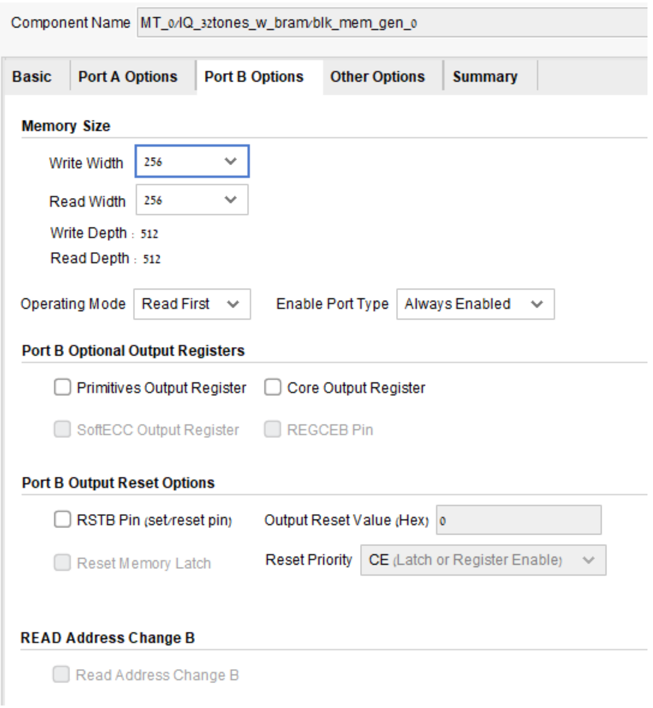


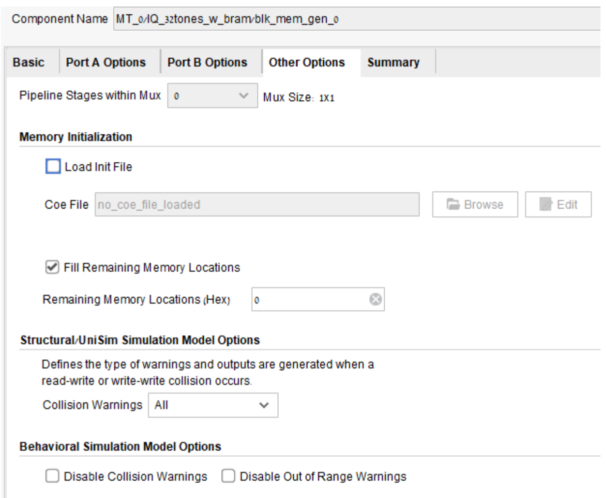
INSIDE SLICING OF EACH TONE WORD
tone_word1_63to0
tone_word2_127to64
tone_word3_191to128
tone_word4_255to192
Slice
s_axi_tones_aclk
S_AXI_tones
s_axi_tones_aresetn
addrb[15:0]
axi_bram_ctrl_1
AXI BRAM Controller
xlslice_0
blk_mem_gen_0
Block Memory Generator
SHOULD BE [14:0]. IN ADDRESS EDITOR: /MT_x/IQ_32tones_w_bram/axi_bram_ctrl_0 /S_AXI SIZE SHOULD BE 32K
IQ_accum_1
IQ_accum_2
IQ_accum_3
IQ_accum_4
tune_words_1
tune_words_2
tune_words_3
tune_words_4
add_12_I
add_12_Q
add_34_I
add_34_Q
sum_all_I
sum_all_Q
Adder/Subtracter
I_out[15:0]
Dout[63:0]
Q_out[15:0]
T_base[31:0]
clk_in[0:0]
dds_ena[0:0]
last_tone[0:0]
A&B are signed 16b integers, representing summing up all signals comprising I and Q out

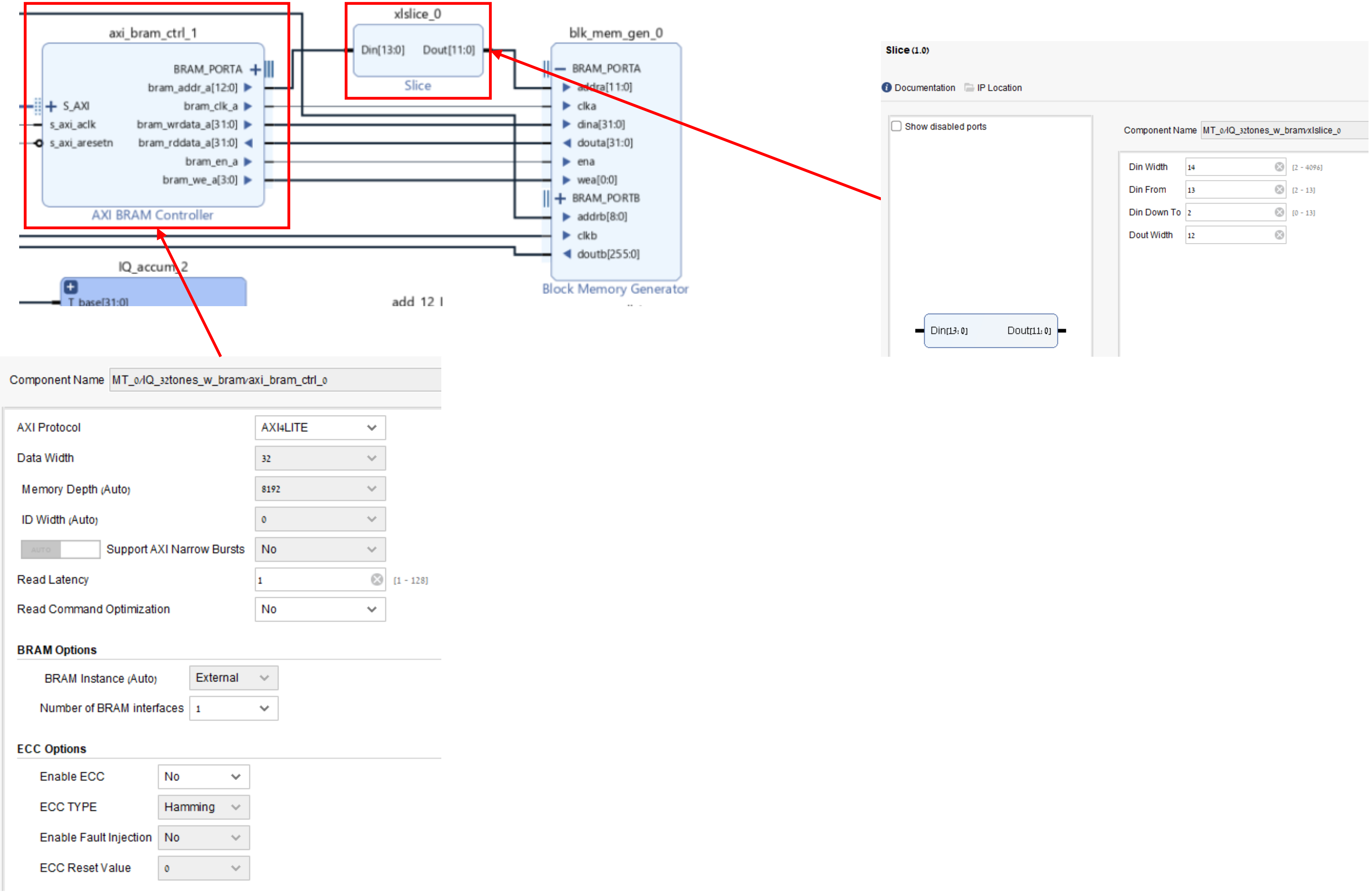

xlslice_0
axi_bram_ctrl_1
blk_mem_gen_0
Din[13:0]
Dout[11:0]
Slice
BRAM_PORTA
bram_addr_a[12:0]
S_AXI
bram_clk_a
s_axi_aclk
bram_wrdata_a[31:0]
s_axi_aresetn
bram_rddata_a[31:0]
bram_en_a
bram_we_a[3:0]
AXI BRAM Controller
BRAM_PORTA
addra[11:0]
clka
dina[31:0]
douta[31:0]
ena
wea[0:0]
BRAM_PORTB
addrb[8:0]
clkb
doutb[255:0]
Block Memory Generator
IQ_accum_2
add_12_I

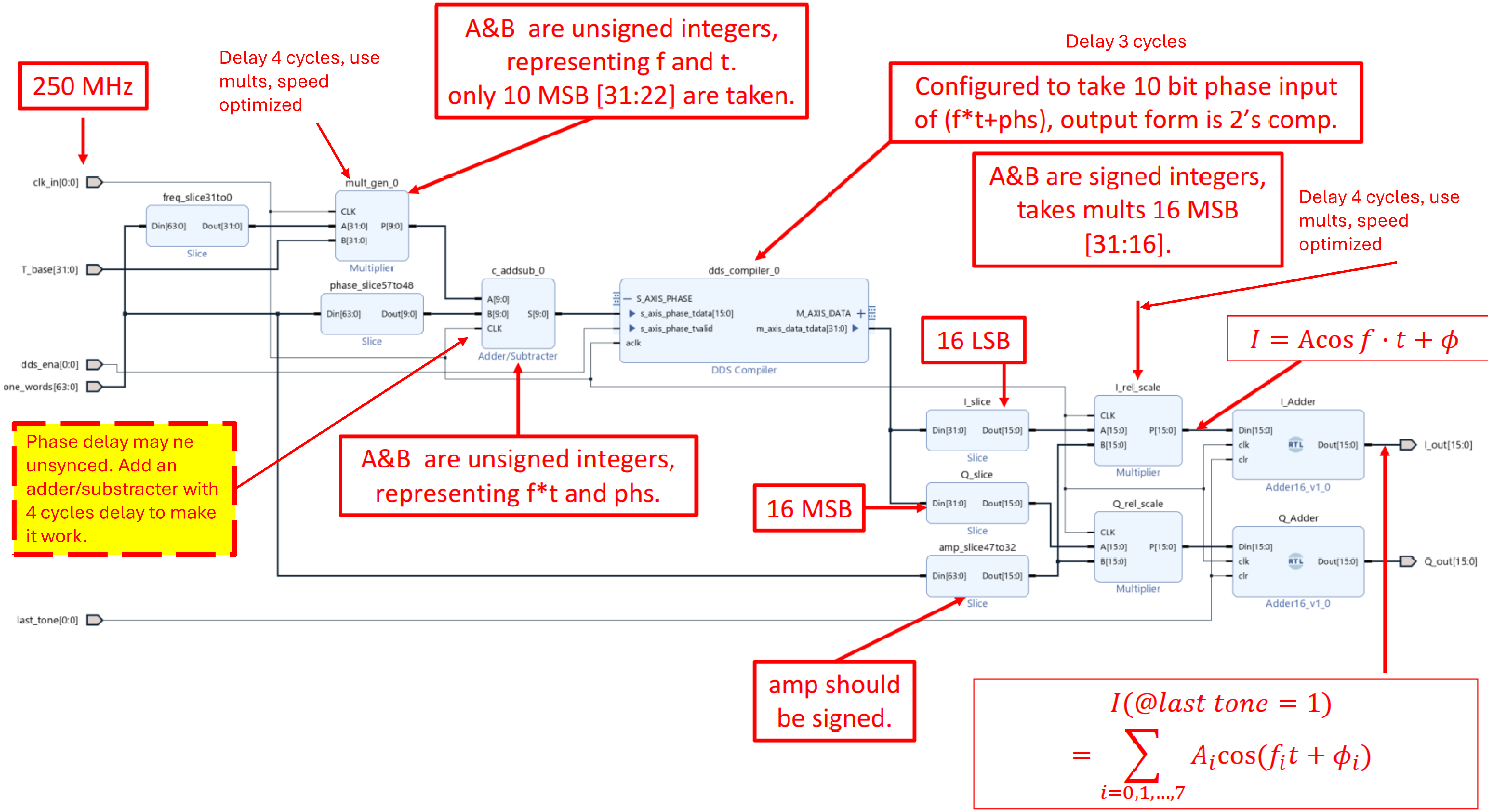

Component Name MT_0/IQ_32tones_w_bram/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Mode Stand Alone
Generate address interface with 32 bits
Memory Type True Dual Port RAM
Common Clock
ECC Options
ECC Type No ECC
Error Injection Pins Single Bit Error Injection
Write Enable
Byte Write Enable
Byte Size (bits) 9
Algorithm Options
Defines the algorithm used to concatenate the block RAM primitives.
Refer datasheet for more information.
Algorithm Minimum Area
Primitive 8kx2


Component Name MT_0/IQ_32tones_w_bram/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Memory Size
Write Width 32
Range: 1 to 4608 (bits)
Read Width 32
Write Depth 4096
Range: 2 to 1048576
Read Depth 4096
Operating Mode Write First
Enable Port Type Use ENA Pin
Port A Optional Output Registers
Primitives Output Register
Core Output Register
SoftECC Input Register
REGCEA Pin
Port A Output Reset Options
RSTA Pin (set/reset pin)
Output Reset Value (Hex) 0
Reset Memory Latch
Reset Priority CE (Latch or Register Enable)
READ Address Change A
Read Address Change A


Component Name MT_0/IQ_32tones_w_bram/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Memory Size
Write Width 256
Read Width 256
Write Depth : 512
Read Depth : 512
Operating Mode Read First
Enable Port Type Always Enabled
Port B Optional Output Registers
Primitives Output Register
Core Output Register
SoftECC Output Register
REGCEB Pin
Port B Output Reset Options
RSTB Pin (set/reset pin)
Output Reset Value (Hex) 0
Reset Memory Latch
Reset Priority CE (Latch or Register Enable)
READ Address Change B
Read Address Change B


Component Name MT_0/IQ_32tones_w_bram/blk_mem_gen_0
Basic
Port A Options
Port B Options
Other Options
Summary
Pipeline Stages within Mux 0
Mux Size: 1x1
Memory Initialization
Load Init File
Coe File no_coe_file_loaded
Browse
Edit
Fill Remaining Memory Locations
Remaining Memory Locations (Hex) 0
Structural/UniSim Simulation Model Options
Defines the type of warnings and outputs are generated when a
read-write or write-write collision occurs.
Collision Warnings All
Behavioral Simulation Model Options
Disable Collision Warnings
Disable Out of Range Warnings


xlslice_0
Din[13:0]
Dout[11:0]
Slice
axi_bram_ctrl_1
BRAM_PORTA
bram_addr_a[12:0]
S_AXI
bram_clk_a
s_axi_aclk
bram_wrdata_a[31:0]
s_axi_aresetn
bram_rddata_a[31:0]
bram_en_a
bram_we_a[3:0]
AXI BRAM Controller
blk_mem_gen_0
BRAM_PORTA
addra[11:0]
clka
dina[31:0]
douta[31:0]
ena
wea[0:0]
BRAM_PORTB
addrb[8:0]
clkb
doutb[255:0]
Block Memory Generator
IQ_accum_2
T_base[31:0]
add_12
Slice (1.0)
Documentation
IP Location
Show disabled ports
Component Name
MT_0/IQ_32tones_w_bram/xlslice_0
Din Width
14
[2 - 4096]
Din From
13
[2 - 13]
Din Down To
2
[0 - 13]
Dout Width
12
Din[13:0]
Dout[11:0]
Component Name
MT_0/IQ_32tones_w_bram/axi_bram_ctrl_0
AXI Protocol
AXI4LITE
Data Width
32
Memory Depth (Auto)
8192
ID Width (Auto)
0
AUTO
Support AXI Narrow Bursts
No
Read Latency
1
[1 - 128]
Read Command Optimization
No
BRAM Options
BRAM Instance (Auto)
External
Number of BRAM interfaces
1
ECC Options
Enable ECC
No
ECC TYPE
Hamming
Enable Fault Injection
No
ECC Reset Value
0



250 MHz
Delay 4 cycles, use mults, speed optimized
A&B are unsigned integers, representing f and t. only 10 MSB [31:22] are taken.
Delay 3 cycles
Configured to take 10 bit phase input of (f*t+phs), output form is 2's comp.
A&B are signed integers, takes mults 16 MSB [31:16].
Delay 4 cycles, use mults, speed optimized
clk_in[0:0]
T_base[31:0]
dds_ena[0:0]
one_words[63:0]
last_tone[0:0]
freq_slice31to0
Din[63:0]
Dout[31:0]
Slice
mult_gen_0
CLK
A[31:0]
B[31:0]
P[9:0]
Multiplier
phase_slice57to48
Din[63:0]
Dout[9:0]
Slice
c_addsub_0
A[9:0]
B[9:0]
CLK
S[9:0]
Adder/Subtracter
dds_compiler_0
S_AXIS_PHASE
s_axis_phase_tdata[15:0]
s_axis_phase_tvalid
aclk
M_AXIS_DATA
m_axis_data_tdata[31:0]
DDS Compiler
16 LSB
$I = A\cos f \cdot t + \phi$
I_slice
Din[31:0]
Dout[15:0]
Slice
Q_slice
Din[31:0]
Dout[15:0]
Slice
amp_slice47to32
Din[63:0]
Dout[15:0]
Slice
I_rel_scale
CLK
A[15:0]
B[15:0]
P[15:0]
Multiplier
Q_rel_scale
CLK
A[15:0]
B[15:0]
P[15:0]
Multiplier
I_Adder
Din[15:0]
clk
clr
RTL
Dout[15:0]
Adder16_v1_0
Q_Adder
Din[15:0]
clk
clr
RTL
Dout[15:0]
Adder16_v1_0
I_out[15:0]
Q_out[15:0]
Phase delay may ne unsynced. Add an adder/substracter with 4 cycles delay to make it work.
A&B are unsigned integers, representing f*t and phs.
16 MSB
amp should be signed.
$I(@last\ tone = 1) = \sum_{i=0,1,\ldots,7} A_i \cos(f_i t + \phi_i)$